

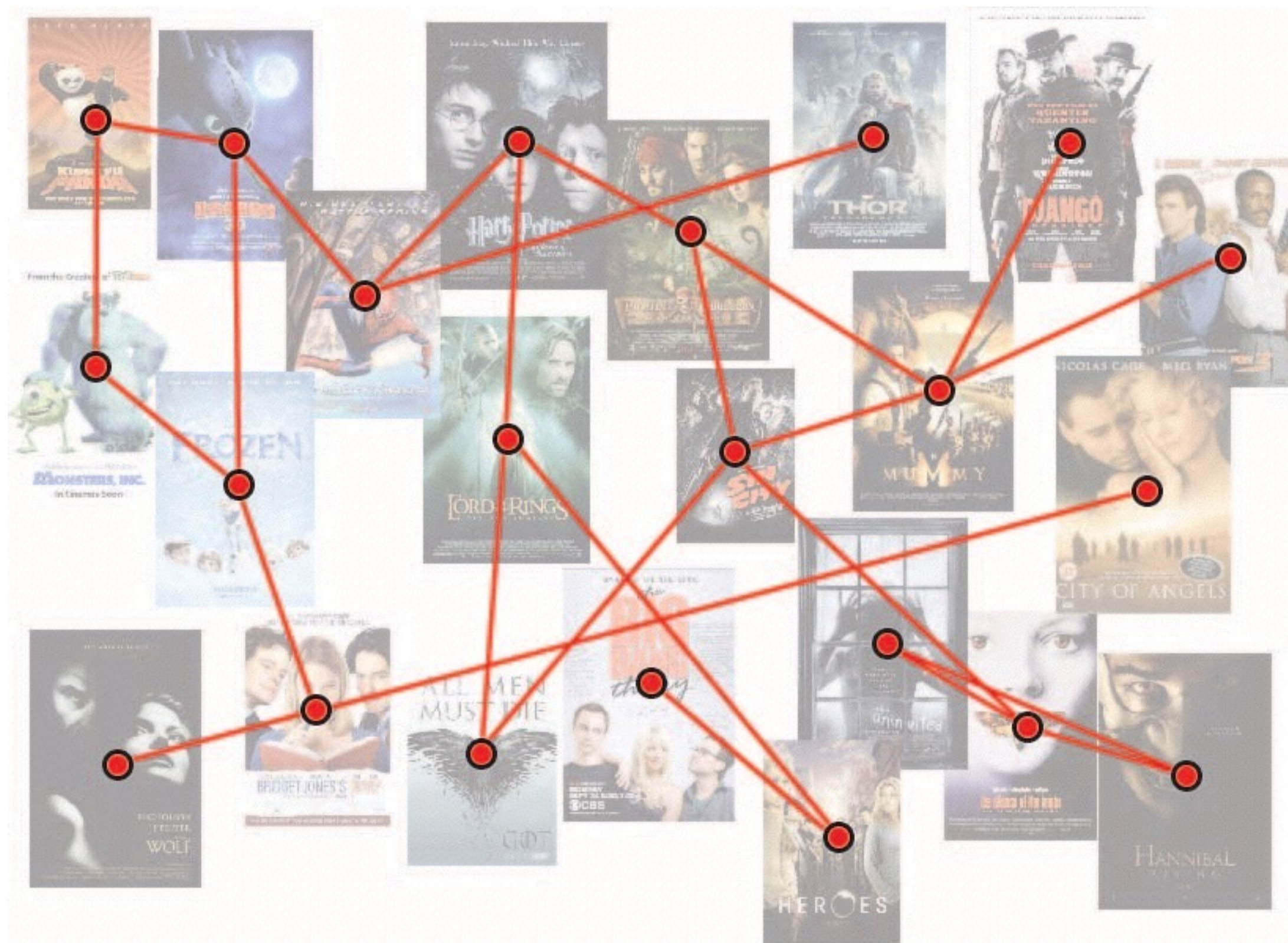


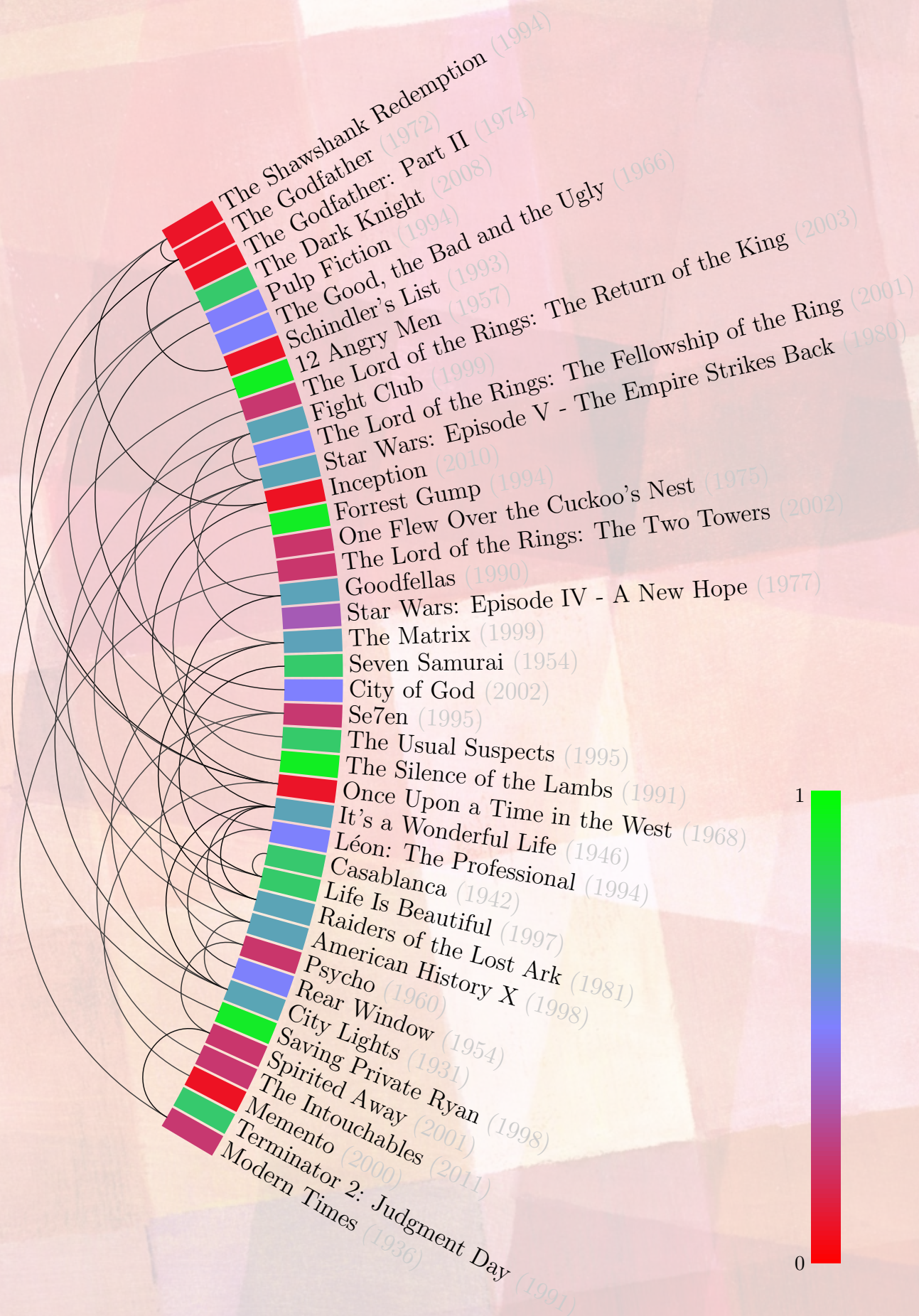
GRAPH BANDITS

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Michal Valko, SequeL, Inria Lille - Nord Europe









Example of a graph bandit problem

movie recommendation

- recommend movies to a **single** user
- **goal:** maximise the sum of the ratings (minimise regret)
- good prediction after just a few steps

$$T \ll N$$

- extra information
 - ratings are **smooth** on a graph
- main question: can we learn **faster**?



#actions

#rounds

$$R_T = \mathcal{O} \left(\sqrt{NT} \right)$$

3,761,592

Problem: Too many actions!

LEARNING FASTER

#actions

#rounds

$$R_T = \mathcal{O} \left(\sqrt{NT} \right)$$

- ▶ Arm independence is too strong and unnecessary
- ▶ Replace N with something much smaller
 - ▶ problem/instance/data dependent
 - ▶ example: linear bandits N to D
- ▶ In this talk: **Graph Bandits!**
 - ▶ sequential problems where **actions** are **nodes** on a graph
 - ▶ find strategies that replace N with a **smaller** graph-dependent quantity

#dimensions



JOINT WORK WITH...



Alexandra Carpentier
Universität Potsdam



Branislav Kveton
Adobe Research



Gergely Neu
Universitat Pompeu Fabra



Manjesh Hanawal
Boston University



Rémi Munos
Google DeepMind



Shipra Agrawal
Columbia University



Tomáš Kocák
Sequel, Inria Lille
graduating in 2016



Venkatesh Saligrama
Boston University

GRAPH BANDITS: GENERAL SETUP

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Every round t the learner

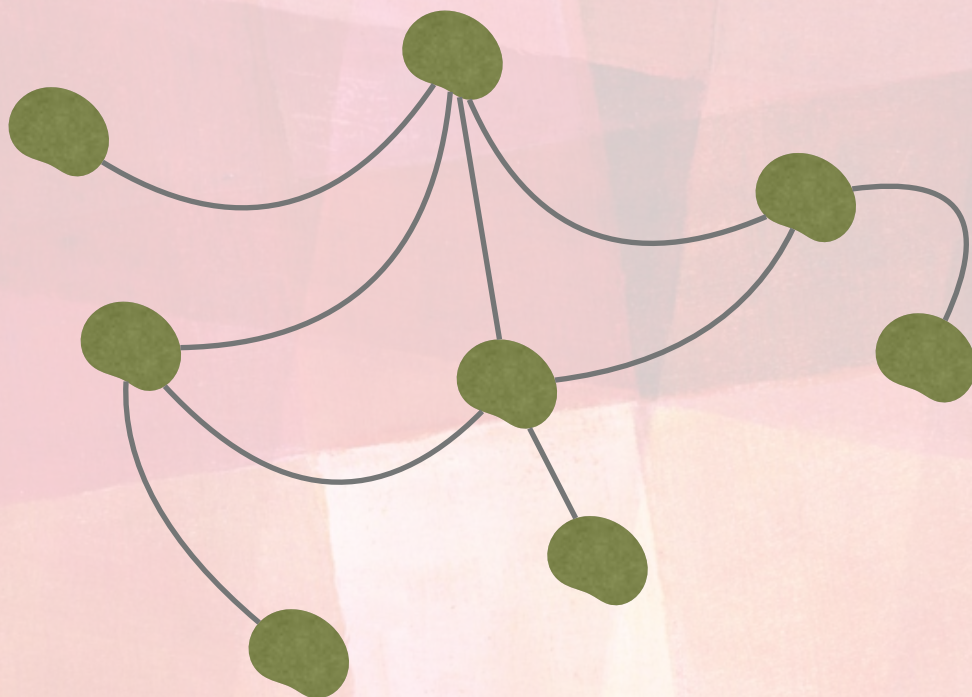
- ▶ picks a node $I_t \in [N]$
- ▶ incurs a loss ℓ_{t,I_t}
- ▶ optional feedback

The performance is total expected regret

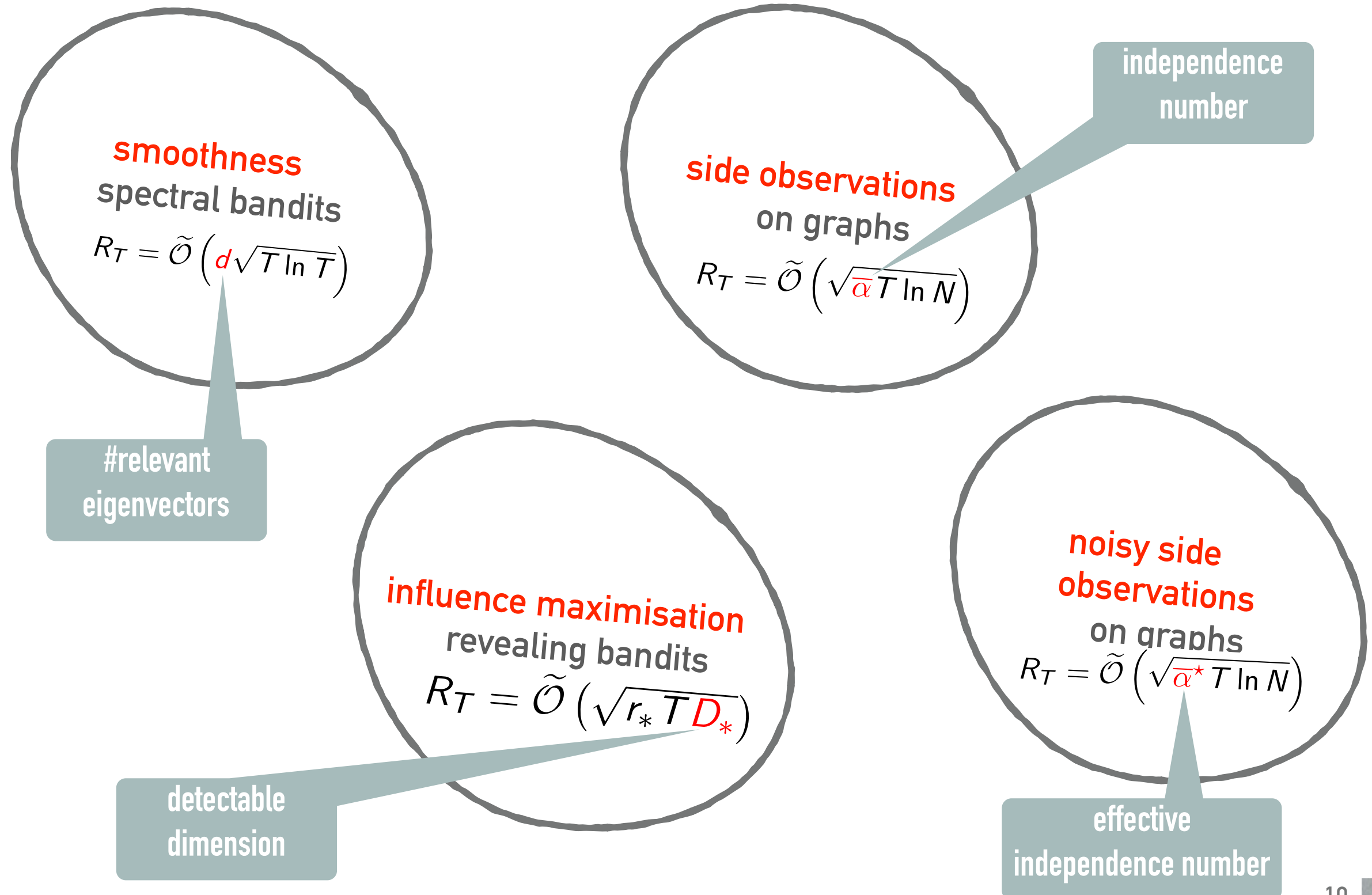
$$R_T = \max_{i \in [N]} \mathbb{E} \left[\sum_{t=1}^T (\ell_{t,I_t} - \ell_{t,i}) \right]$$

1. loss
2. feedback
3. guarantees

Specific setups differ in



SPECIFIC **GRAPH** BANDIT SETTINGS



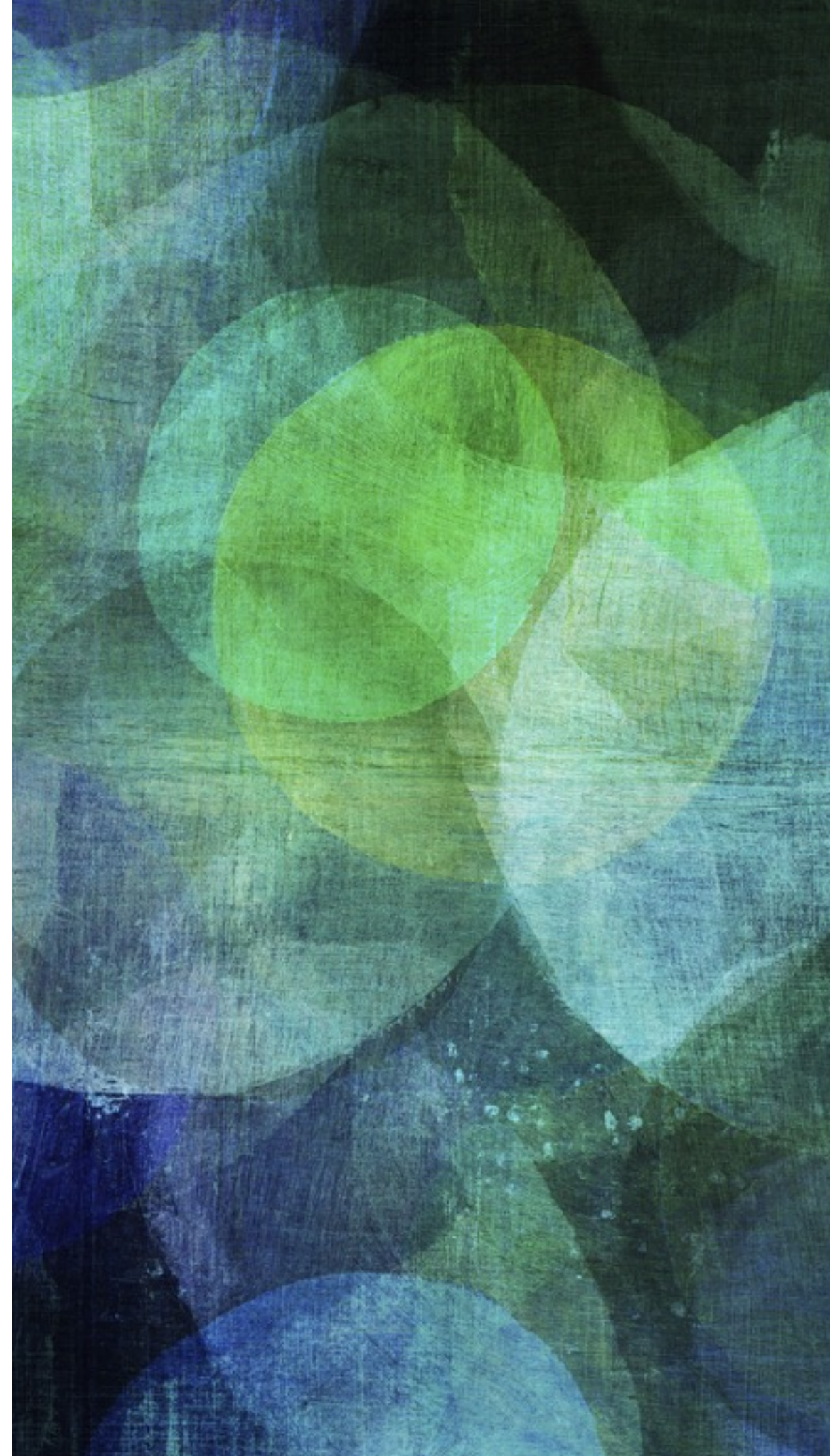
MV, Munos, Kveton, Kocák: **Spectral Bandits for Smooth Graph Functions**, ICML 2014

Kocák, MV, Munos, Agrawal: **Spectral Thompson Sampling**, AAAI 2014

Hanawal, Saligrama, MV, Munos: **Cheap Bandits**, ICML 2015

SPECTRAL BANDITS

.....
exploiting smoothness of
rewards on graphs



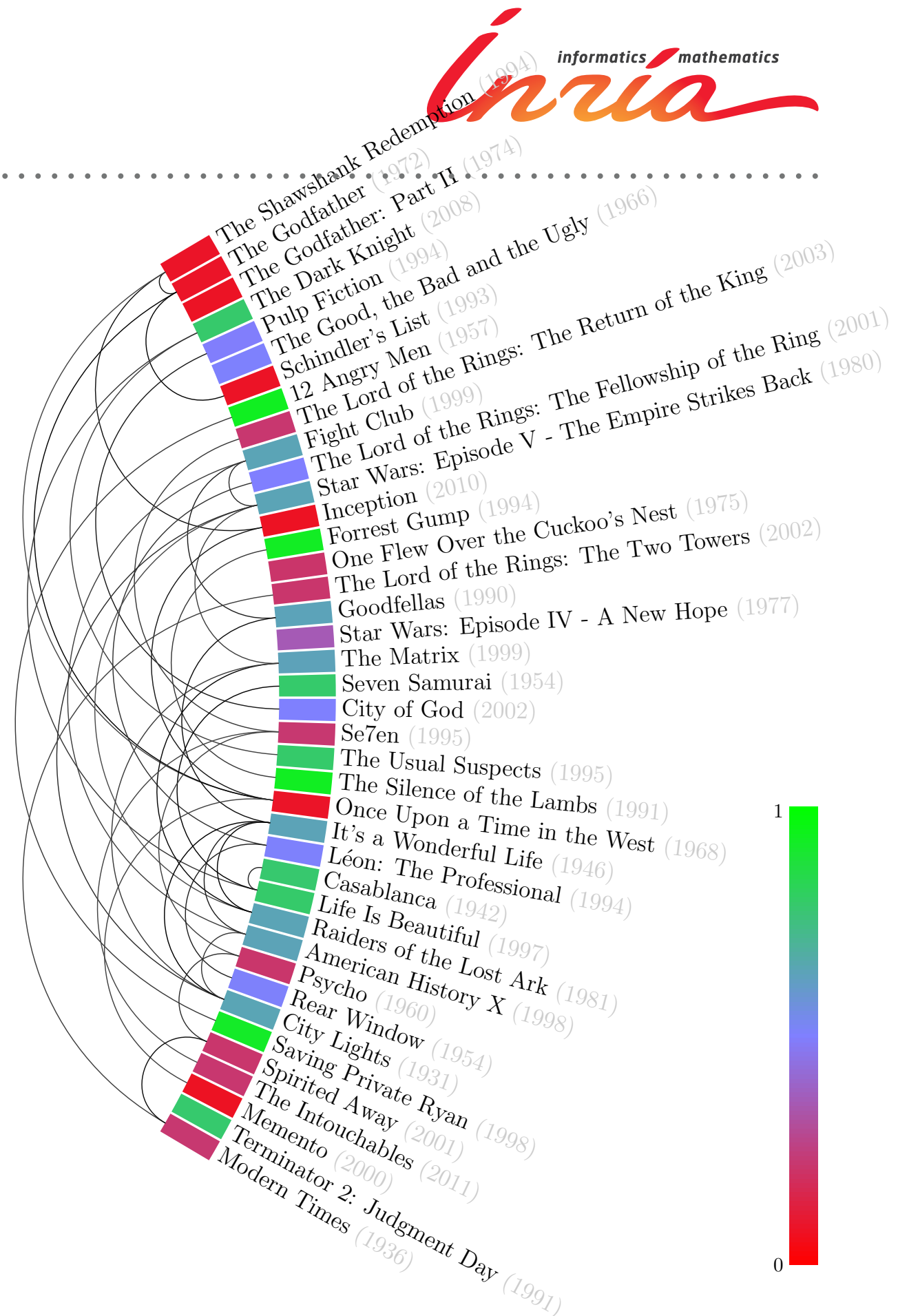
SPECTRAL BANDITS

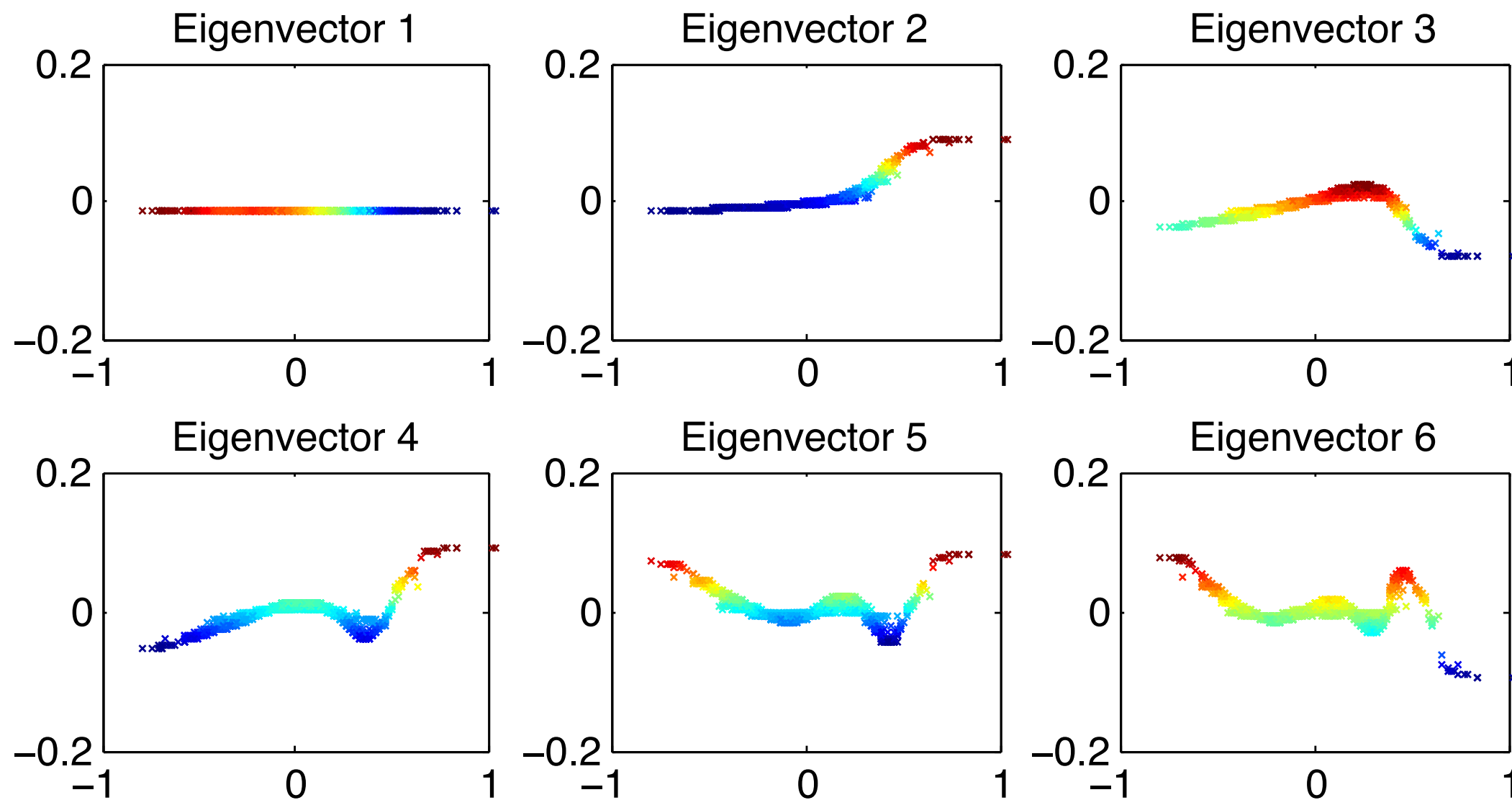
Assumptions

- ▶ Unknown reward function $f : V(G) \rightarrow \mathbb{R}$.
- ▶ Function f is **smooth** on a graph.
- ▶ Neighboring movies $\Rightarrow$ similar preferences.
- ▶ Similar preferences $\nRightarrow$ neighboring movies.

Desiderata

An algorithm useful in the case $T \ll N!$





Eigenvectors from the Flixster data corresponding to the smallest few eigenvalues of the graph Laplacian projected onto the first principal component of data. Colors indicate the values.

- ▶ $\mathbf{f} = (f_1, \dots, f_N)^\top$: Vector of function values.
- ▶ Let $\mathbf{L} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ be the eigendecomposition of the Laplacian.
 - ▶ Diagonal matrix $\mathbf{\Lambda}$ whose diagonal entries are eigenvalues of $\mathbf{L}$.
 - ▶ Columns of $\mathbf{Q}$ are eigenvectors of $\mathbf{L}$.
 - ▶ Columns of $\mathbf{Q}$ form a basis.
- ▶ α^* : Unique vector such that $\mathbf{Q}\alpha^* = \mathbf{f}$

$$\frac{1}{2} \sum_{i,j \leq n} w_{i,j} (f_i - f_j)^2$$

Note: $\mathbf{Q}^\top \mathbf{f} = \alpha^*$

$$S_G(\mathbf{f}) = \mathbf{f}^\top \mathbf{L} \mathbf{f} = \mathbf{f}^\top \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{f} = \alpha^{*\top} \mathbf{\Lambda} \alpha^* = \|\alpha^*\|_{\mathbf{\Lambda}}^2 = \sum_{i=1}^N \lambda_i (\alpha_i^*)^2$$

Smoothness and regularization: Small value of

(a) $S_G(\mathbf{f})$ (b) $\mathbf{\Lambda}$ norm of α^* (c) α_i^* for large λ_i

Learning setting for a bandit algorithm π

- ▶ In each time t step choose a node $\pi(t)$.
- ▶ the $\pi(t)$ -th **row** $\mathbf{x}_{\pi(t)}$ of the matrix $\mathbf{Q}$ corresponds to the arm $\pi(t)$.
- ▶ Obtain noisy reward $r_t = \mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^* + \varepsilon_t$. **Note:** $\mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^* = f_{\pi(t)}$
 - ▶ ε_t is R -sub-Gaussian noise. $\forall \xi \in \mathbb{R}, \mathbb{E}[e^{\xi \varepsilon_t}] \leq \exp(\xi^2 R^2 / 2)$
- ▶ Minimize cumulative regret

$$R_T = T \max_a (\mathbf{x}_a^\top \boldsymbol{\alpha}^*) - \sum_{t=1}^T \mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^*.$$

Can we just use linear bandits?

LINEAR VS. SPECTRAL BANDITS

▶ Linear bandit algorithms

▶ LinUCB

(Li et al., 2010)

- ▶ Regret bound $\approx D\sqrt{T \ln T}$

▶ LinearTS

(Agrawal and Goyal, 2013)

- ▶ Regret bound $\approx D\sqrt{T \ln N}$

Note: D is ambient dimension, in our case N , length of x_i .

Number of actions, e.g., all possible movies → **HUGE!**

▶ Spectral bandit algorithms

▶ SpectralUCB

(Valko et al., ICML 2014)

- ▶ Regret bound $\approx d\sqrt{T \ln T}$
- ▶ Operations per step: $D^2 N$

▶ SpectralTS

(Kocák et al., AAAI 2014)

- ▶ Regret bound $\approx d\sqrt{T \ln N}$
- ▶ Operations per step: $D^2 + DN$

Note: d is **effective dimension**, usually much smaller than D .

- ▶ **Effective dimension:** Largest d such that

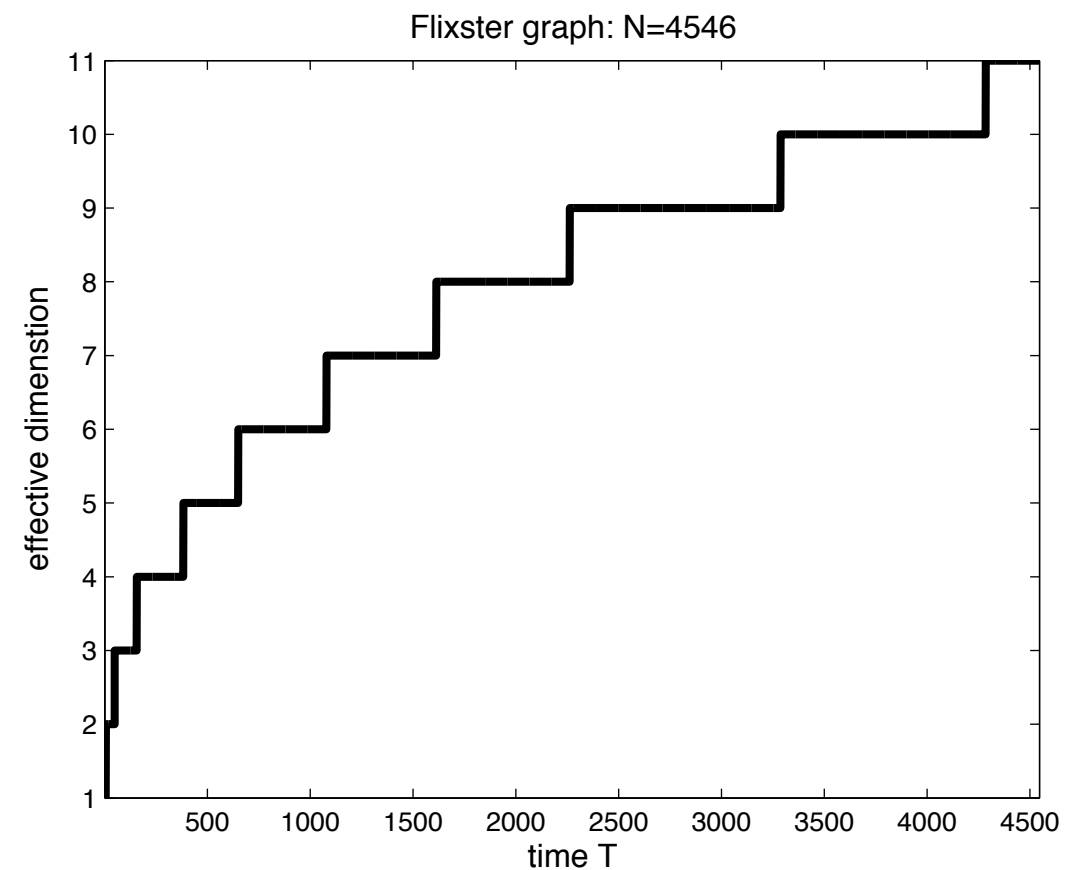
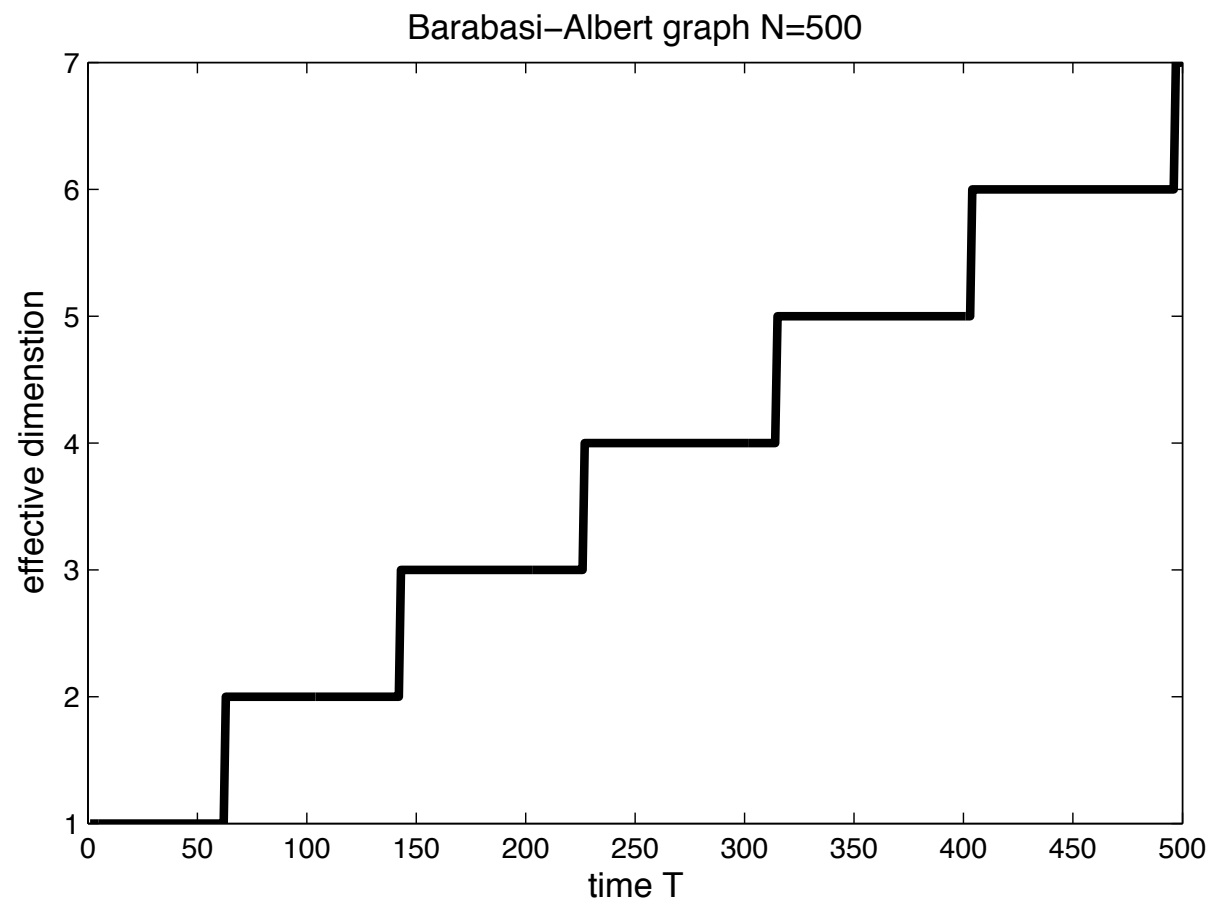
$$(d - 1)\lambda_d \leq \frac{T}{\log(1 + T/\lambda)}.$$

- ▶ Function of time horizon and graph properties
- ▶ λ_i : i -th smallest eigenvalue of $\mathbf{\Lambda}$.
- ▶ λ : Regularization parameter of the algorithm.

Properties:

- ▶ d is small when the coefficients λ_i grow rapidly above time.
- ▶ d is related to the number of “non-negligible” dimensions.
- ▶ Usually d is much smaller than D in real world graphs.
- ▶ Can be computed beforehand.

SPECTRAL BANDITS – EFFECTIVE DIMENSION



$$d \ll D$$

Note: In our setting $T < N = D$.

Given a vector of weights α , we define its $\mathbf{\Lambda}$ norm as

$$\|\alpha\|_{\mathbf{\Lambda}} = \sqrt{\sum_{k=1}^N \lambda_k \alpha_k^2} = \sqrt{\alpha^\top \mathbf{\Lambda} \alpha},$$

and fit the ratings r_v with a (regularized) least-squares estimate

$$\hat{\alpha}_t = \arg \min_{\alpha} \left(\sum_{v=1}^t [\langle \mathbf{x}_v, \alpha \rangle - r_v]^2 + \|\alpha\|_{\mathbf{\Lambda}}^2 \right).$$

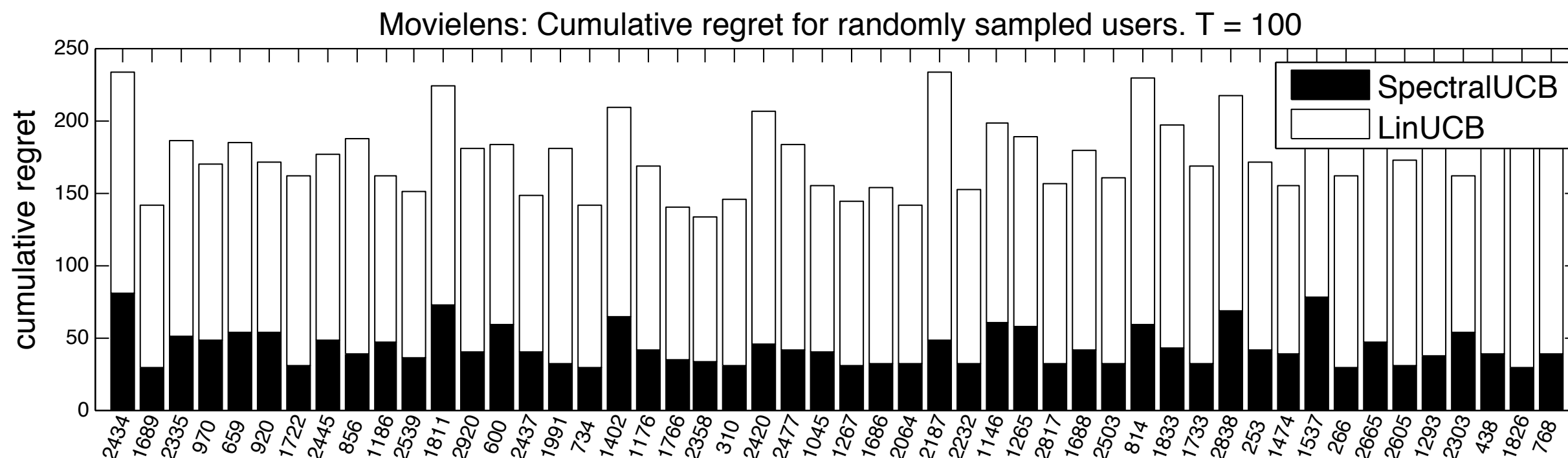
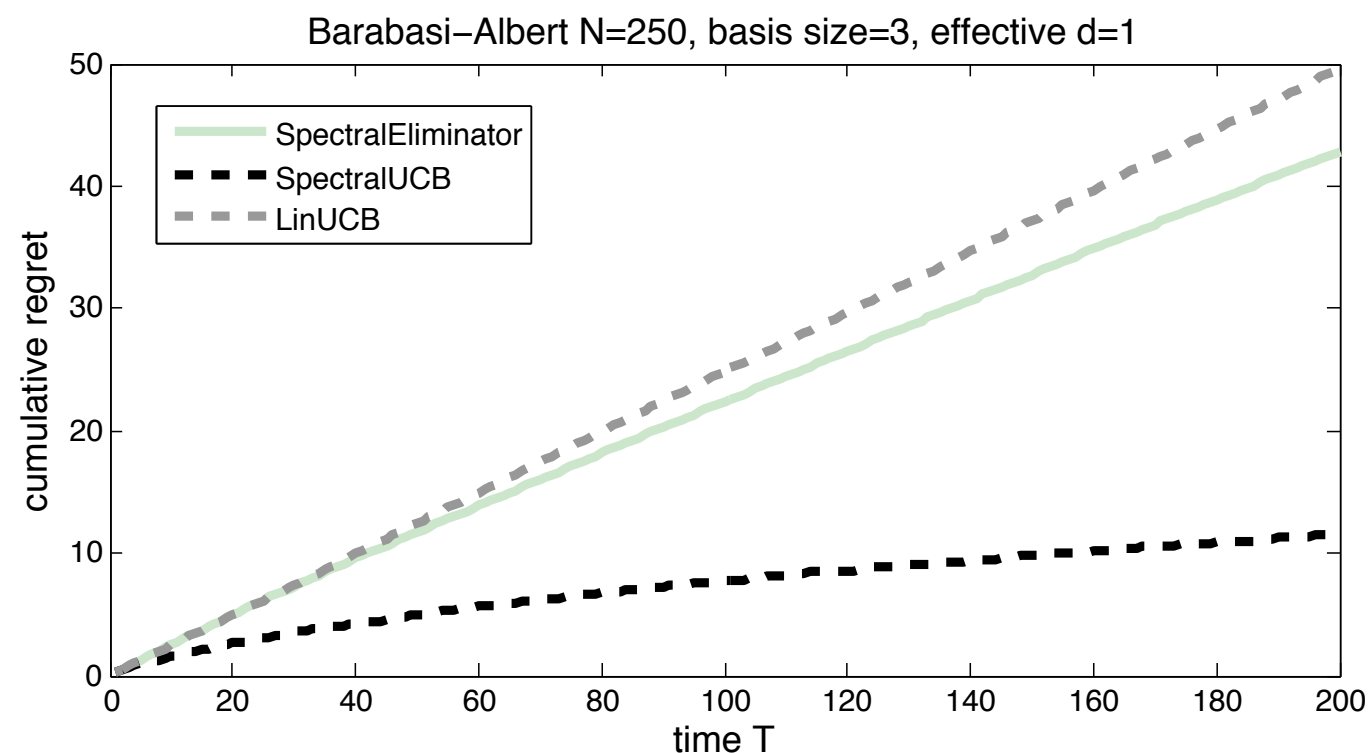
$\|\alpha\|_{\mathbf{\Lambda}}$ is a penalty for non-smooth combinations of eigenvectors.

- ▶ d : Effective dimension.
- ▶ λ : Minimal eigenvalue of $\mathbf{\Lambda} = \mathbf{\Lambda}_L + \lambda \mathbf{I}$.
- ▶ C : Smoothness upper bound, $\|\alpha^*\|_{\mathbf{\Lambda}} \leq C$.
- ▶ $\mathbf{x}_i^\top \alpha^* \in [-1, 1]$ for all i .

The **cumulative regret** R_T of **SpectralUCB** is with probability $1 - \delta$ bounded as

$$R_T \leq \left(8R \sqrt{d \ln \frac{\lambda + T}{\lambda}} + 2 \ln \frac{1}{\delta} + 4C + 4 \right) \sqrt{dT \ln \frac{\lambda + T}{\lambda}}.$$

$$R_T \approx d \sqrt{T \ln T}$$

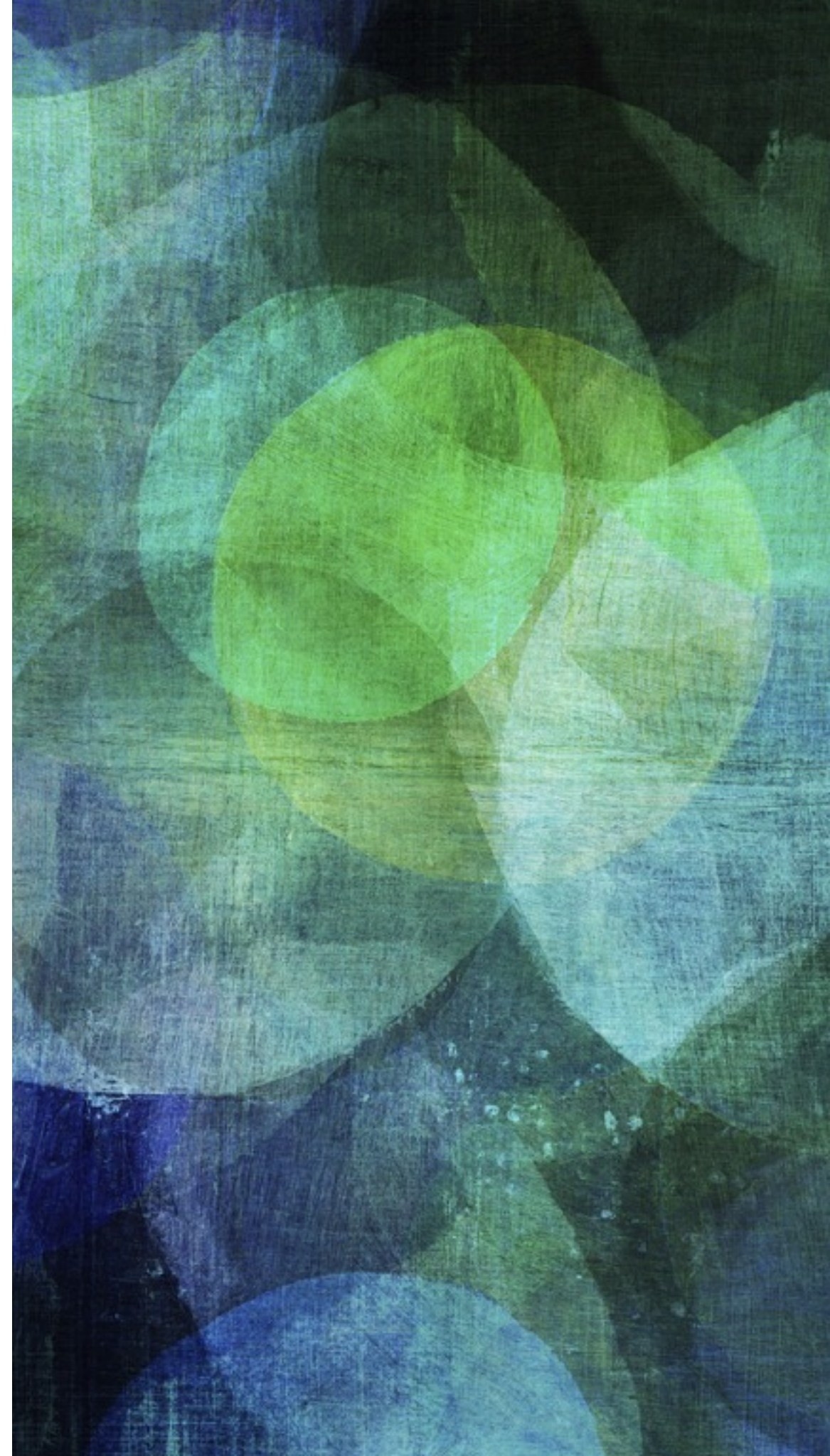


Kocák, Neu, MV, Munos: Efficient learning by implicit exploration
in bandit problems with side observations, NIPS 2014

Kocák, Neu, MV: Online learning with Erdős-Rényi side-observation
graphs, UAI 2016 (to appear)

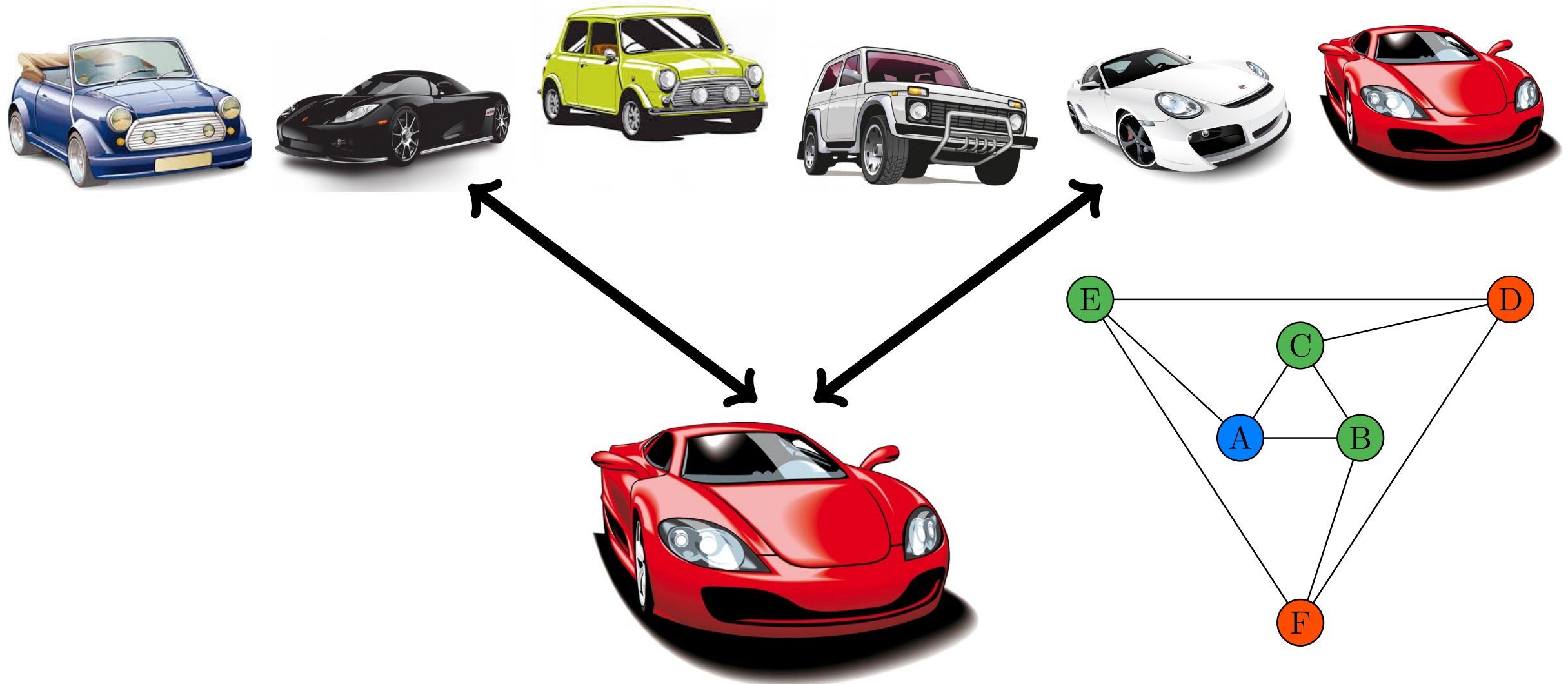
GRAPH BANDITS WITH SIDE OBSERVATIONS

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exploiting **free** observations from
neighbouring nodes



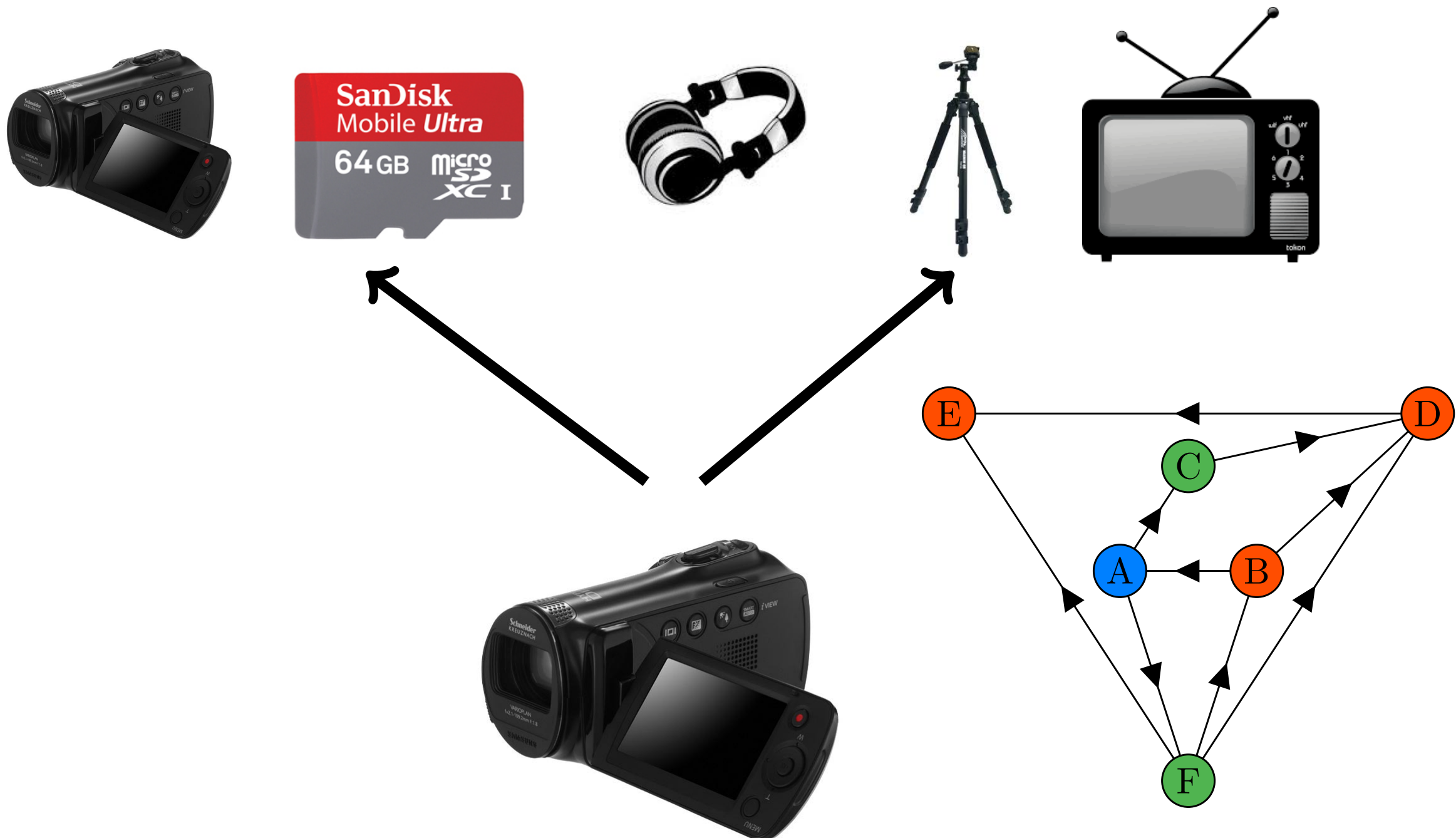
SIDE OBSERVATIONS: UNDIRECTED

Example 1: undirected observations



SIDE OBSERVATIONS: DIRECTED

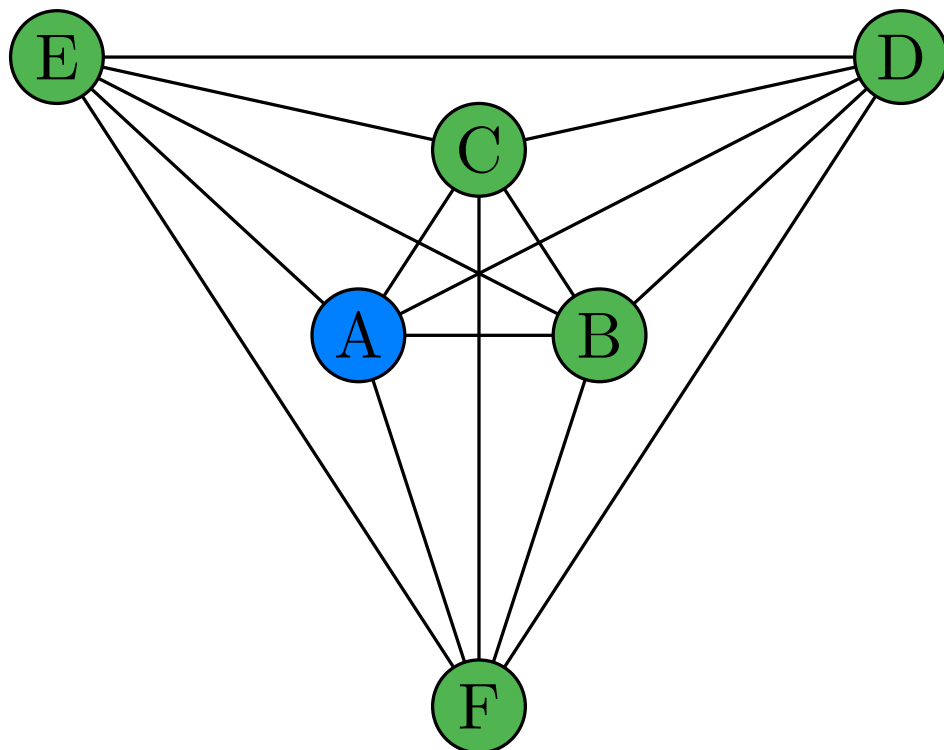
Example 2: Directed observation



SIDE OBSERVATIONS – AN INTERMEDIATE GAME

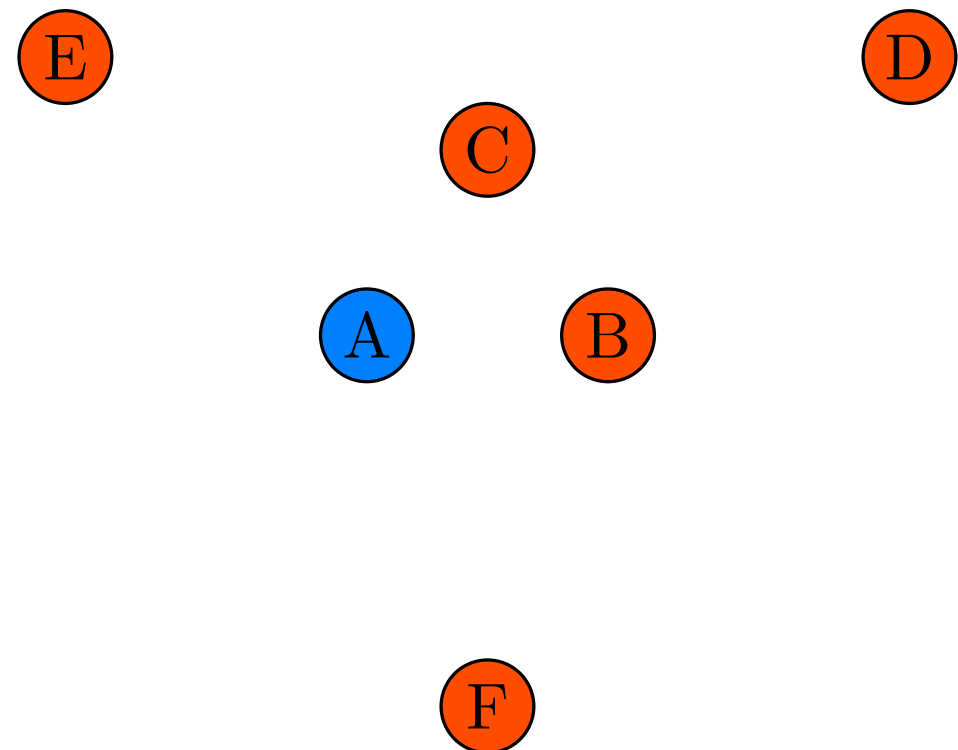
Full Information setting

- ▶ Pick an action (e.g. action A)
- ▶ Observe losses of all actions
- ▶ $R_T = \tilde{O}(\sqrt{T})$



Bandit setting

- ▶ Pick an action (e.g. action A)
- ▶ Observe loss of a chosen action
- ▶ $R_T = \tilde{O}(\sqrt{NT})$



UNDIRECTED GRAPHS

Side observation (Undirected case)

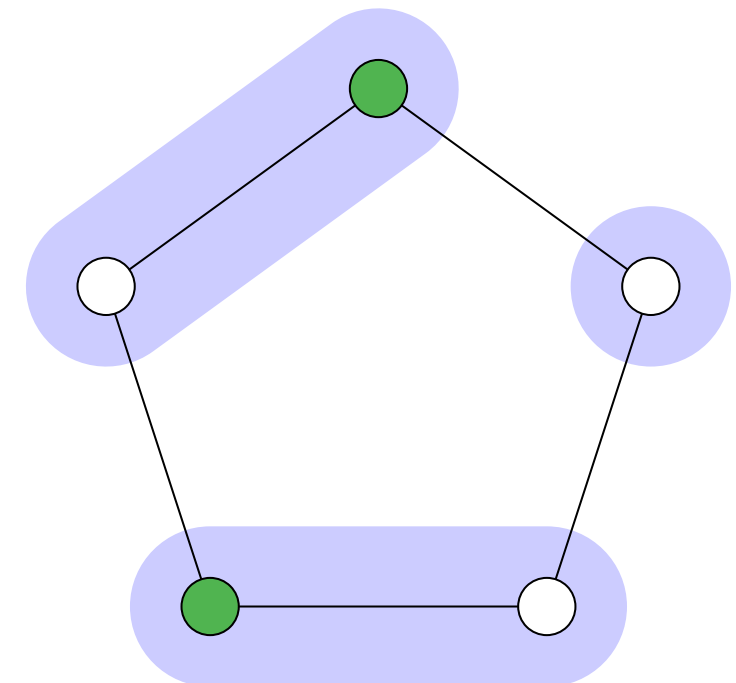
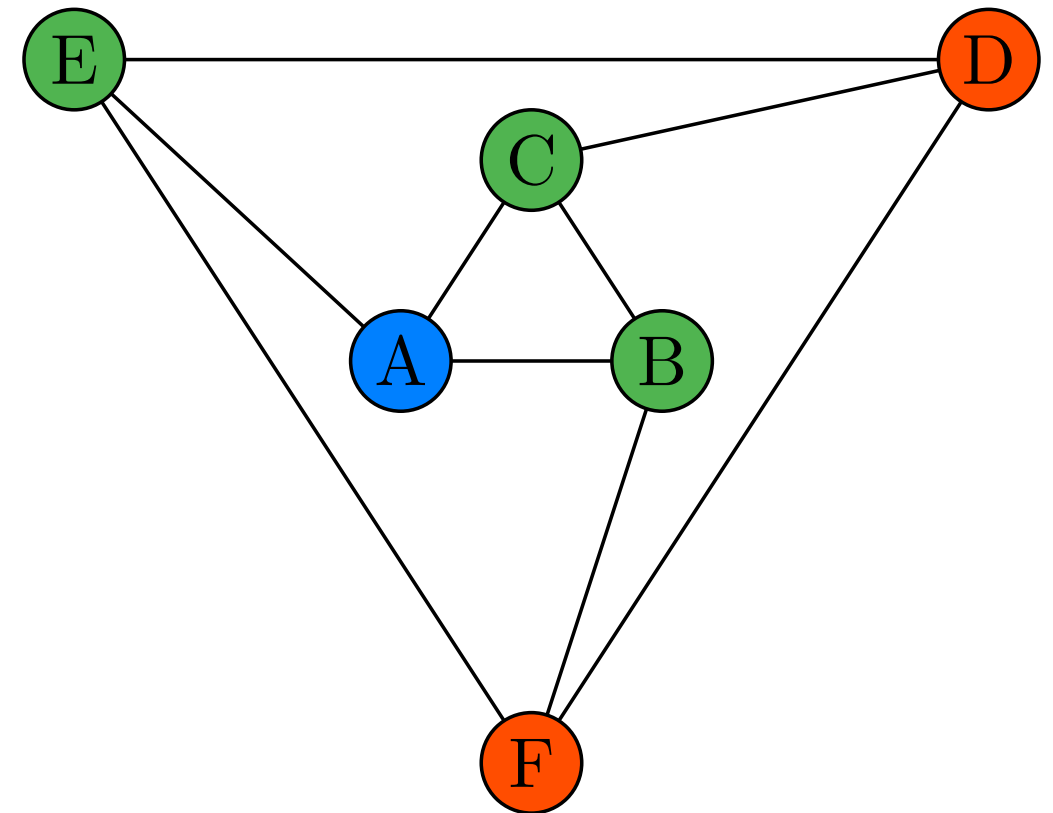
- ▶ Pick an action (e.g. action A)
- ▶ Observe losses of neighbors

Mannor and Shamir (ELP algorithm)

- ▶ Need to know the graph
- ▶ Clique decomposition (c cliques)
- ▶ $R_T = \tilde{O}(\sqrt{cT})$

Alon, Cesa-Bianchi, Gentile, Mansour

- ▶ No need to know the graph
- ▶ Independence set of α actions
- ▶ $R_T = \tilde{O}(\sqrt{\alpha T})$



Side observation (**Directed case**)

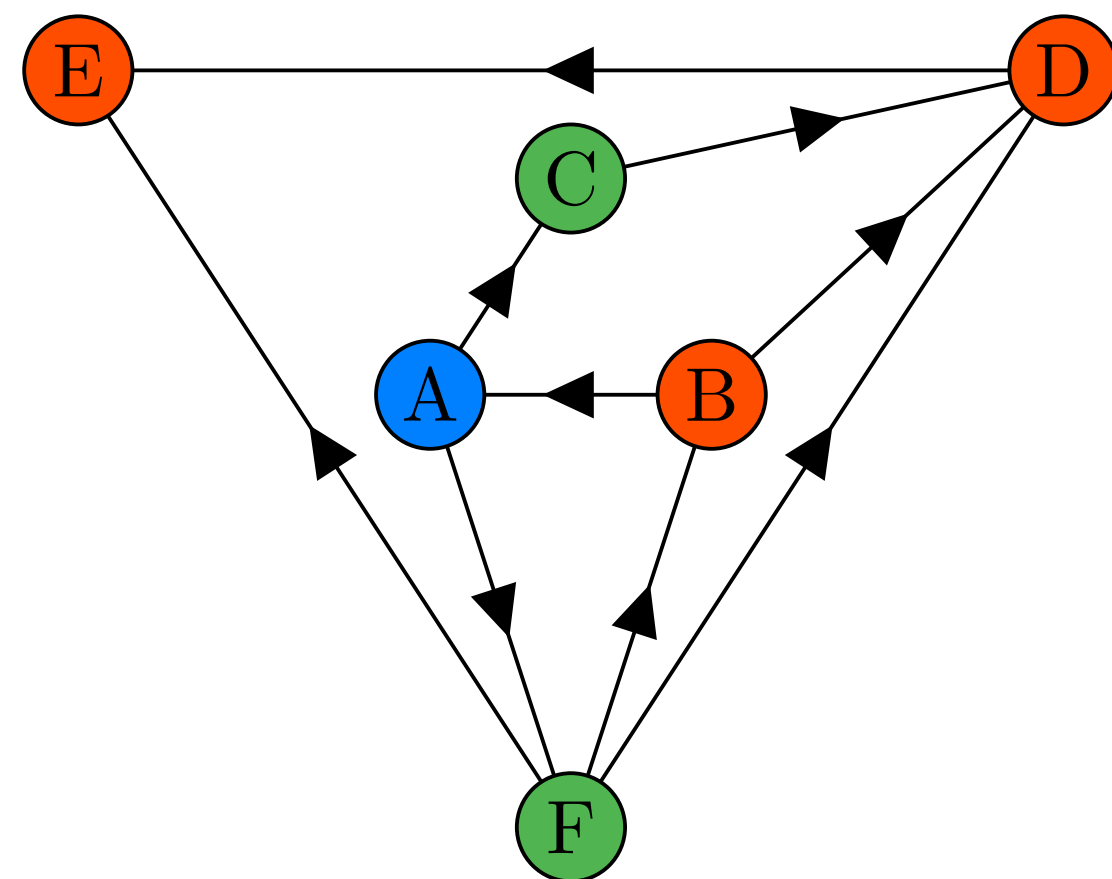
- ▶ Pick an action (e.g. action A)
- ▶ Observe losses of neighbors

Alon, Cesa-Bianchi, Gentile, Mansour

- ▶ **Exp3-DOM**
- ▶ Need to know graph
- ▶ Need to find dominating set
- ▶ $R_T = \tilde{O}(\sqrt{\alpha T})$

Exp3-IX - Kocák et. al

- ▶ No need to know graph
- ▶ $R_T = \tilde{O}(\sqrt{\alpha T})$



Algorithm 1 EXP3-IX

```

1: Input: Set of actions  $\mathcal{S} = [d]$ ,
2:   parameters  $\gamma_t \in (0, 1)$ ,  $\eta_t > 0$  for  $t \in [T]$ .
3: for  $t = 1$  to  $T$  do
4:    $w_{t,i} \leftarrow (1/d) \exp(-\eta_t \hat{L}_{t-1,i})$  for  $i \in [d]$ 
5:   An adversary privately chooses losses  $\ell_{t,i}$ 
     for  $i \in [d]$  and generates a graph  $G_t$ 
6:    $W_t \leftarrow \sum_{i=1}^d w_{t,i}$ 
7:    $p_{t,i} \leftarrow w_{t,i}/W_t$ 
8:   Choose  $I_t \sim \mathbf{p}_t = (p_{t,1}, \dots, p_{t,d})$ 
9:   Observe graph  $G_t$ 
10:  Observe pairs  $\{i, \ell_{t,i}\}$  for  $(I_t \rightarrow i) \in G_t$ 
11:   $o_{t,i} \leftarrow \sum_{(j \rightarrow i) \in G_t} p_{t,j}$  for  $i \in [d]$ 
12:   $\hat{\ell}_{t,i} \leftarrow \frac{\ell_{t,i}}{o_{t,i} + \gamma_t} \mathbb{1}_{\{(I_t \rightarrow i) \in G_t\}}$  for  $i \in [d]$ 
13: end for

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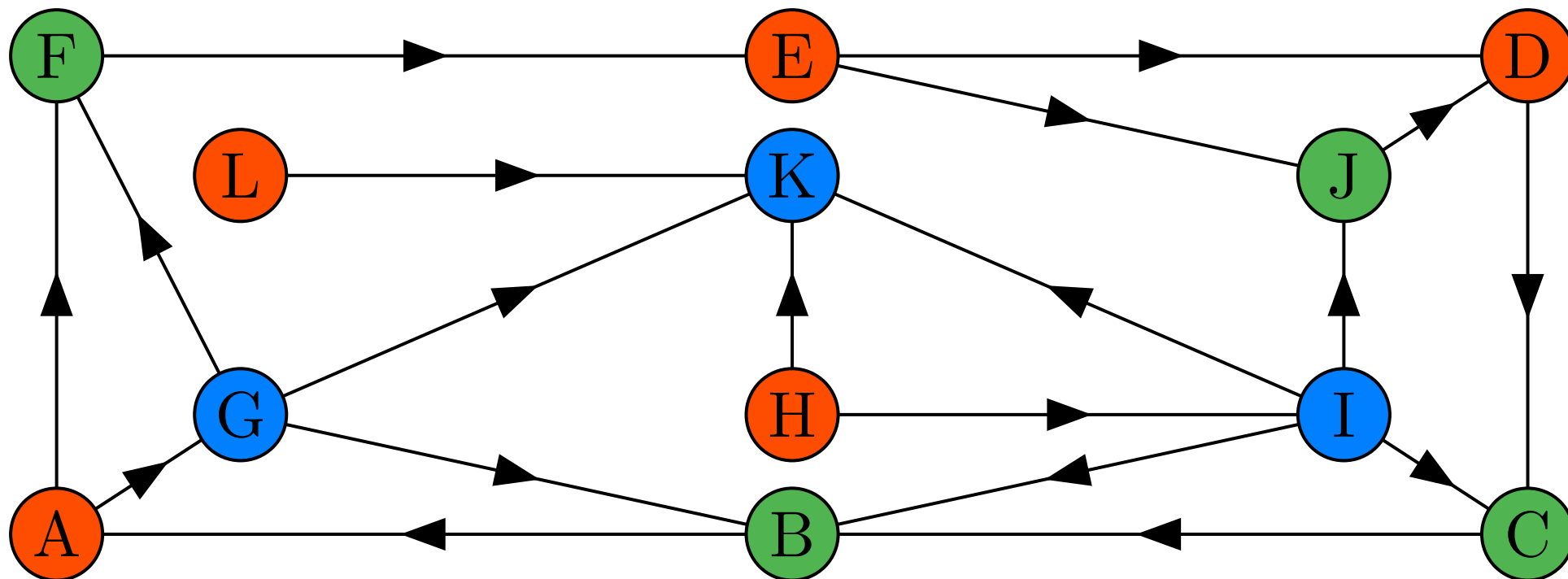
Benefits of the implicit exploration

- ▶ no need to know the graph before
- ▶ no need to estimate dominating set
- ▶ no need for doubling trick
- ▶ no need for aggregation

$$R_T = \tilde{O} \left(\sqrt{\alpha T \ln N} \right)$$

Optimistic bias for the loss estimates

$$\mathbb{E}[\hat{\ell}_{t,i}] = \frac{\ell_{t,i}}{o_{t,i} + \gamma} o_{t,i} + 0(1 - o_{t,i}) = \ell_{t,i} - \ell_{t,i} \frac{\gamma}{o_{t,i} + \gamma} \leq \ell_{t,i}$$



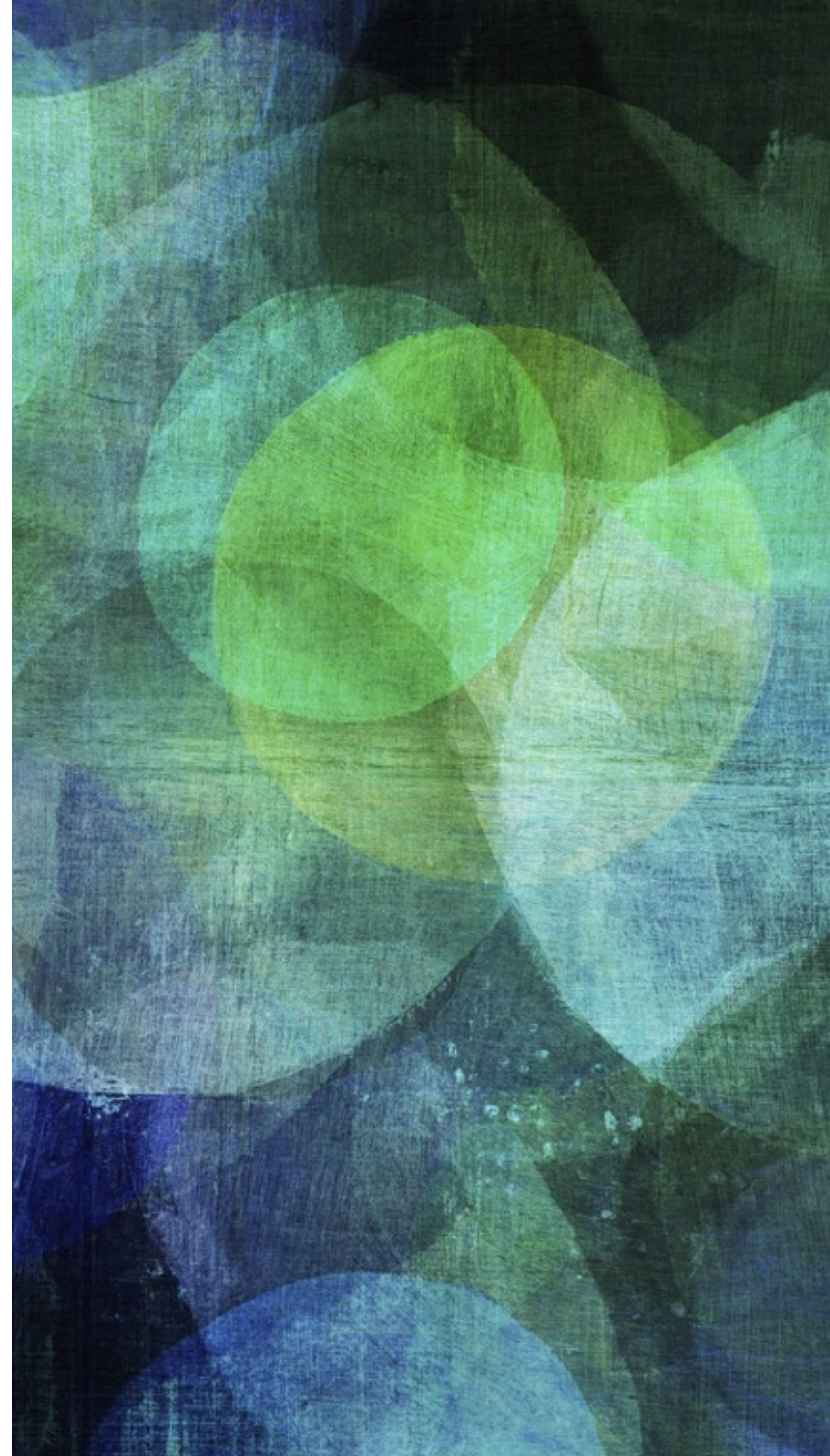
- ▶ Play action $\mathbf{V}_t \in S \subset \{0, 1\}^N$, $\|\mathbf{v}\|_1 \leq m$ from all $\mathbf{v} \in S$
- ▶ Obtain losses $\mathbf{V}_t^\top \ell_t$
- ▶ Observe additional losses according to the graph

$$R_T = \tilde{O} \left(m^{3/2} \sqrt{\sum_{t=1}^T \alpha_t} \right) = \tilde{O} \left(m^{3/2} \sqrt{\alpha T} \right)$$

Reducing the knowledge of the graph...

INFLUENCE MAXIMISATION

.....
looking for the influential nodes
while exploring the graph

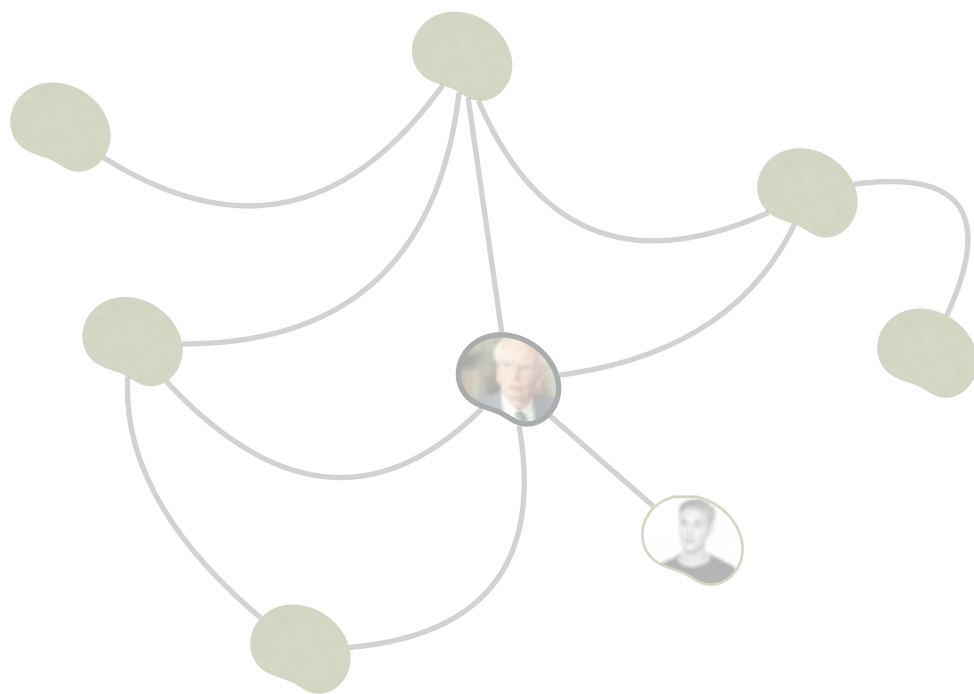


REVEALING BANDITS FOR **LOCAL** INFLUENCE

Unknown $\mathbf{M} = (p_{i,j})_{i,j}$ symmetric matrix of influences

In each time step $t = 1, \dots, T$

- ▶ learners picks a node k_t
- ▶ set $S_{k_t,t}$ of influenced nodes is *revealed*



Select influential people = Find the strategy maximising

$$L_T = \sum_{t=1}^T |S_{k_t,t}|$$

The number of expected influences of node k is by definition

$$r_k = \mathbb{E}[|S_{k,t}|] = \sum_{j \leq N} p_{k,j}$$

Oracle strategy always selects the best

$$k^* = \arg \max_k \mathbb{E} \left[\sum_{t=1}^T |S_{k,t}| \right] = \arg \max_k Tr_k$$

Expected regret of any adaptive, non-oracle strategy **unaware** of $\mathbf{M}$

$$\mathbb{E}[R_T] = \mathbb{E}[L_T^*] - \mathbb{E}[L_T]$$

Ignoring the structure again? The best we can do is $\tilde{O}(\sqrt{r_* T N})$

We aim to do better: $R_T = \tilde{O}(\sqrt{r_* T D_*})$

reward of the
best node

D_* - detectable dimension dependent on T and the structure

- ▶ *good case*: star-shaped graph can have $D_* = 1$
- ▶ *bad case*: a graph with many small cliques.
- ▶ *the worst case*: all nodes are disconnected except 2

Idea of the algorithm:

- ▶ *exploration phase*: sample randomly to find out $\approx D_*$ nodes
- ▶ *bandit case*: use any bandit algorithm on these nodes

EMPIRICAL RESULTS

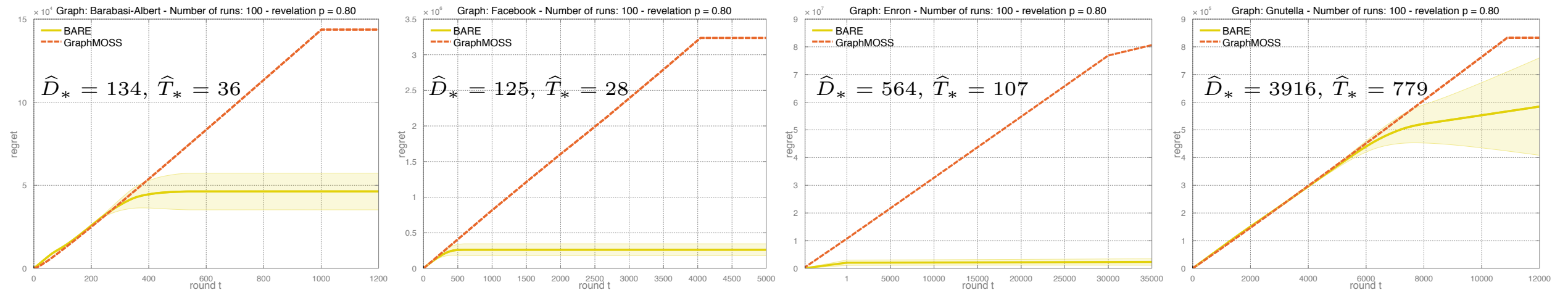


Figure 1: *Left: Barabási-Albert. Middle left: Facebook. Middle right: Enron. Right: Gnutella.*

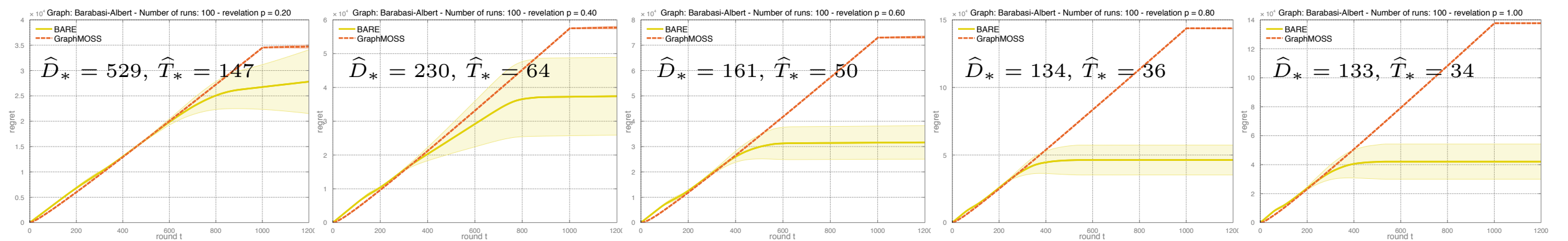


Figure 2: Barabási-Albert model with varying p between 0.2 and 1

► Enron and Facebook vs. Gnutella (decentralised)

CONCLUSION AND NEW DIRECTIONS

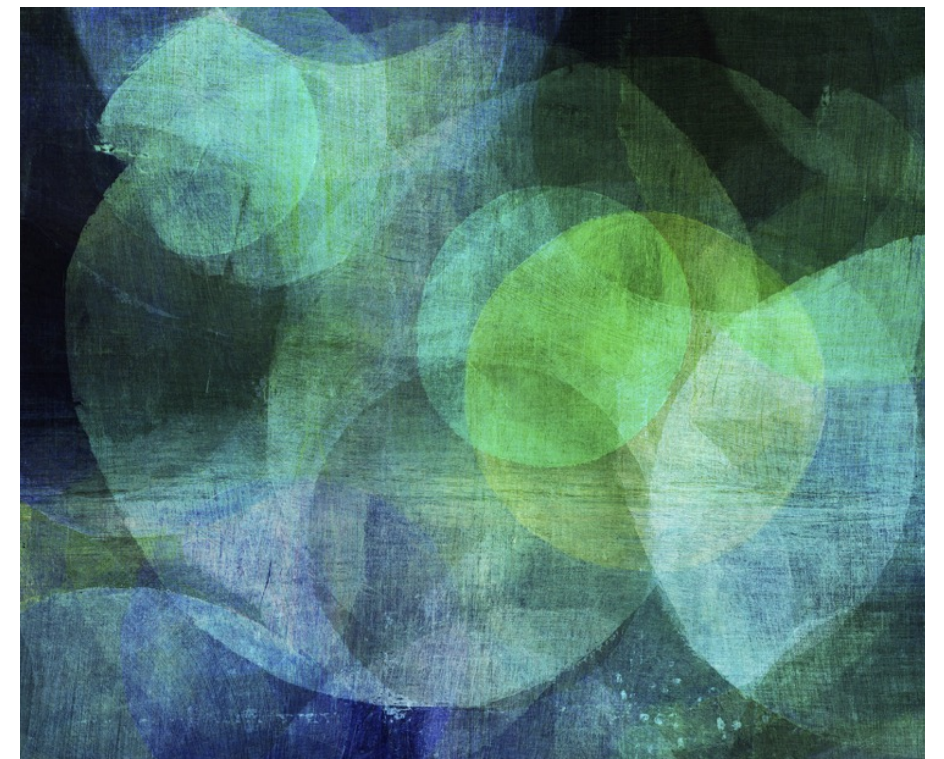
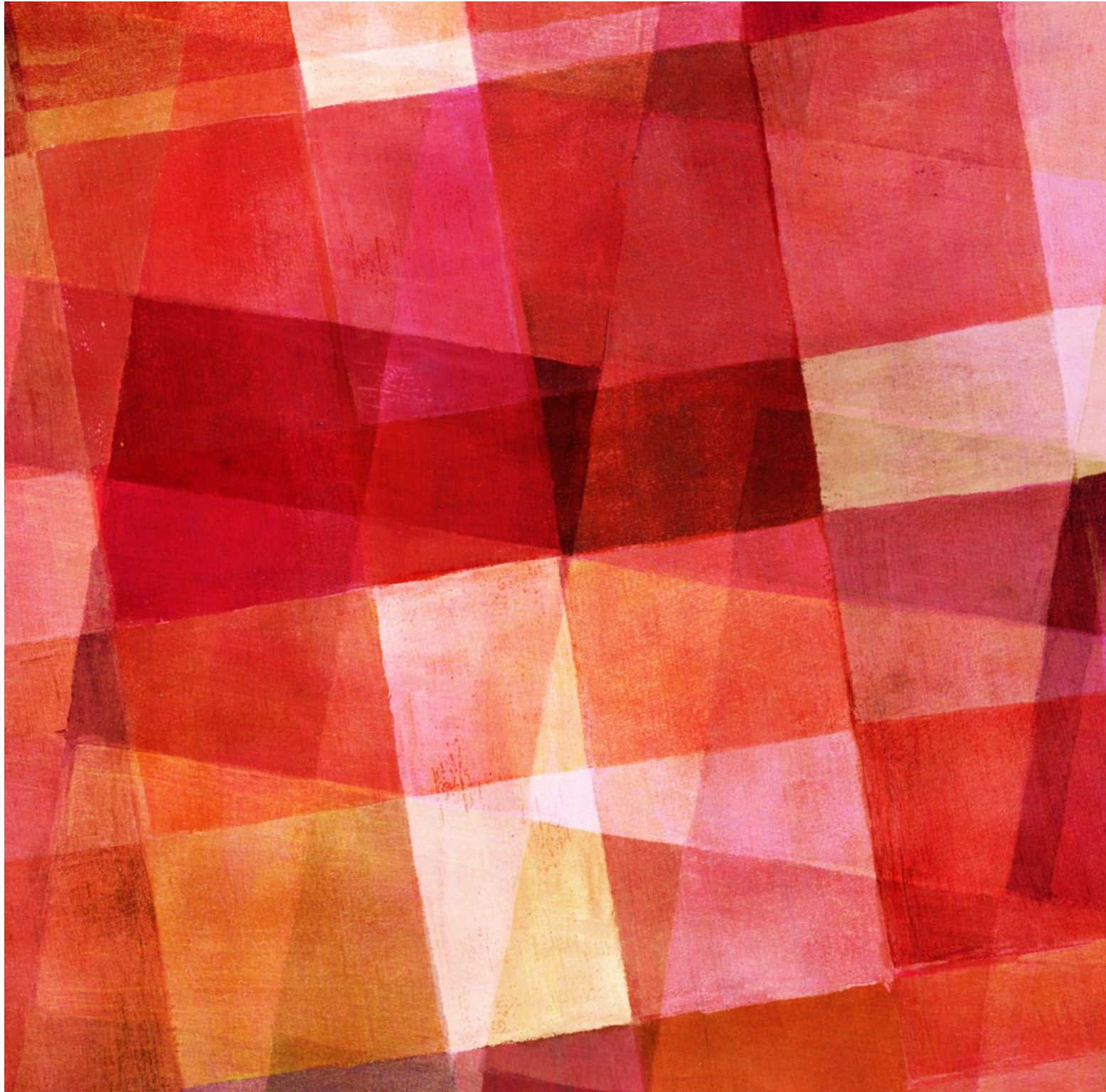
Graph Bandits

- ▶ specific way of exploiting the problem **structure** to learn **faster**
- ▶ different settings
 - ▶ smooth rewards - **spectral** bandits, **cheap** bandits
 - ▶ (perfect, noisy) side observations - **informed** bandits
 - ▶ influence maximisation - **revealing** bandits

New directions

- ▶ other graph generators (BA, ...), **Erdős-Rényi** side observations (to appear at UAI 25-29. 6. 2016)
- ▶ learning (with) communities - SBM
- ▶ bandit learning global influence models (ongoing)
- ▶ crawling strategies
- ▶ reducing assumption on graph knowledge





Data Driven Approach to Networks and Language 2016, ENS de Lyon, France

Graph Bandits: Michal Valko, SequeL, Inria Lille - Nord Europe, michal.valko@inria.fr

<http://researchers.lille.inria.fr/~valko/hp/>